MA51900 Homework 1
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Due 3SEP2026

Handout 1, Question 4 (David)

Six cookies are distributed completely at random, one by one, to six kids. What is the probability that Two kids get two cookies each, and another three get one each.

Answer

The probability of this event happening is 0 since it is outside of the sample space. For two kids to get two cookies each, and another three to each get one would require seven cookies. However, this problem requires that a total of six cookies get distributed. Therefore, this event does not exist in the sample space, and therefore has 0 probability of occuring.

Handout 1, Question 8 (Andre)

If n balls are placed completely at random into n cells, find the probability that exactly one cell remains empty. Next, find the limit of this probability as n tends to infinity rigorously.

Handout 1, Question 10 (David)

(Feller, Vol. I). A group of 100 boys and 100 girls are divided at random into two groups of 100 people. Find the probability that each group has an equal number of boys and girls. Replace 100 by $2N$ and derive a general formula.

Answer

The sample space more than 100, because it must include different combinations of which boys and girls go into each group. Think of it like a coin toss per boy and girl. This is 2^{200}. So, what is number of EXACTLY 100 H tosses out of 200 total tosses? $\binom{200}{100}$. That is the sample space.

$$\|\Omega\| = \binom{200}{100}$$

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Our event A is the event in which exactly 50 boys are in Group A. The count of this is $\binom{100}{50}$. However, we also must choose WHICH 50 girls end up in group A. This also ends up with a count of $\binom{100}{50}$. Therefore, the event count is $\binom{100}{50}^2$.

$$|A| = \binom{100}{50}^2$$

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$$P(A) = \frac{\binom{100}{50}^2}{\binom{200}{100}} = \frac{(\frac{100!}{50!50!})^2}{\frac{200!}{100!100!}} = \frac{(100!)^4}{200!(50!)^4} = \frac{(51 * 52 * ... * 100)^4}{200!} = 0.1128$$

.

The probability is 11.28%.

We can generalize this as:

$$P(A) = \frac{\binom{2N}{N}^2}{\binom{4N}{2N}}$$

Handout 1, Question 29 (Andre)

Six people are to be seated around a circular table.

a)

What is the total number of ways that they can be seated?

b)

What is the total number of ways that they can be seated if two people X and Y refuse to sit next to each other?

Handout 1, Question 31 (David)

The letters in INDIANA are randomly rearranged. What is the probability that on rearrangement, the letters will still read INDIANA?

Answer

We treat each letter as unique for the sake of random combinations. The number of possible combinations for 7 letters is 7!.

$$\|\Omega\| = 7! = 5040$$

Event space must account for order AND for repeat letters. We have repeated letters for I, N, and A, with a total of two each. Therefore, these letters can switch while keeping the same word. The number of iterations in which INDIANA stays the same word, then, is found by $2 \cdot 2 \cdot 2$.

Therefore, we have:

$$\|A\| = 2 \cdot 2 \cdot 2 = 8$$

$$P(A) = \frac{2 \cdot 2 \cdot 2}{7!} = \frac{8}{5040} = \frac{1}{630} = 0.001587$$

The probability of randomly arranging the letters and still getting INDIANA is 0.1587%.

Parzen, Page 5, Question 1.3 (Andre)

Describe how you would explain to a layman the meaning of the following statement:

> An insurance company is not gambling with its clients because it knows with sufficient accuracy what will happen to every thousand or ten thousand or a million people even when the company cannot tell what will happen to any individual among them.

Parzen, Page 16, Question 4.1 (David)

An experiment consists of drawing 3 radio tubes from a lot and testing them for some characteristic of interest. If a tube is defective, assign the letter D to it. If a tube is good, assign the letter G to it. A drawing is then described by a 3-tuple, each of whose components is either D or G. For example, (D, G, G) denotes the outcome that the first tube drawn was defective and the remaining 2 were good. Let A_1 denote the event that the first tube drawn was defective, A_2 denote the event that the second tube drawn was defective, and A_3 denote the event that the third tube drawn was defective. Write down the sample description space of the experiment and list all sample descriptions in the events

$$A_1,\ A_2,\ A_3,\ A_1 \cup A_2,\ A_1 \cup A_3,\ A_2 \cup A_3,\ A_1 \cup A_2 \cup A_3,\ A_1A_2,\ A_1A_3,\ A_2A_3,\ A_1A_2A_3.$$

Answer

Parzen, Page 22-23, Question 5.5 (Andre)

Let A and B be 2 events on a probability space. In terms of $P[A]$, $P[B]$, and $P[AB]$, express

(i)

for $k = 0, 1, 2,\ P[\text{exactly } k \text{ of the events, } A \text{ and } B, \text{ occur}]$,

(ii)

for $k = 0, 1, 2,\ P[\text{at least } k \text{ of the events, } A \text{ and } B, \text{ occur}]$,

(iii)

for $k = 0, 1, 2$, P[at most k of the events, A and B, occur],

(iv)

$P[A$ occurs and B does not occur]

Parzen, Page 22-23, Question 5.10 (David)

In a very hotly fought battle in a small war 270 men fought. Of these, 90 lost an eye, 90 lost an arm, and 90 lost a leg; 30 lost both an eye and an arm, 30 lost both and arm and a leg, and 30 lost both a leg and an eye; 10 lost all three. Find, for $k = 0, 1, 2, 3$, the number of men who suffered these injuries:

(i)

exactly k

(ii)

at least k

(iii)

no more than k

Parzen, Page 22-23, Question 5.11 (Andre)

Certain data obtained from a study of a group of 1000 subscribers to a certain magazine relating to their sex, marital status, and education were reported as follows: 312 males, 470 married, 525 college graduates, 42 male college graduates, 147 married college graduates, 86 married males, and 25 married male college graduates. Show that the numbers reported in the various groups are not consistent.